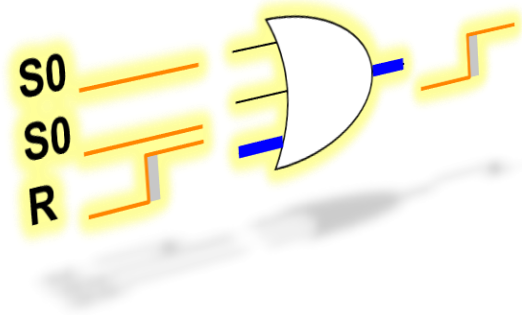


Delay Test

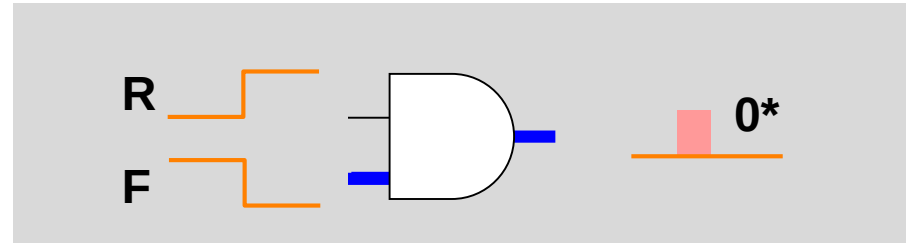
- Introduction and delay fault models
- **Path Delay Fault**
 - ◆ Path Sensitization
 - ◆ Test Generation [Lin 1987]
 - ◆ Fault Simulation
 - * Enumerative method [Smith 1985]
 - * Non-enumerative method [Pomeranz 1994]
- Transition Delay Fault
- Experimental Results* (not in exam)
- Issues of Delay Tests* (not in exam)
- Conclusions



Six-valued Logic

- *Nine-valued* logic simulation for path sensitization (see 9.2)
- *Six-valued* logic for fault simulation
 - ♦ Remove 'X'. assume all inputs are fully specified
 - ♦ Keep 0* and 1*, so *non-robust* sensitization considered

Value	Meaning
S0	Static 0
S1	Static 1
R	Rising
F	Falling
0*	Static-0 hazard
1*	Static-1 hazard
U1	X→1
U0	X→0
XX	X→X



AND	S0	S1	R	F	0*	1*
S0	0	0	0	0	0	0
S1	0	1	R	F	0*	1*
R	0	R	R	0*	0*	R
F	0	F	0*	F	0*	F
0*	0	0*	0*	0*	0*	0*
1*	0	1*	R	F	0*	1*

Path Delay Fault Simulation

① *Enumerative* method

😊 Simple algorithm

😞 Users need to provide fault list

* How? Use **static timing analyzer (STA)** to find long paths

😞 Simulation fault coverage may not be representative

② *Non-enumerative* method

😊 Users don't need to provide fault list

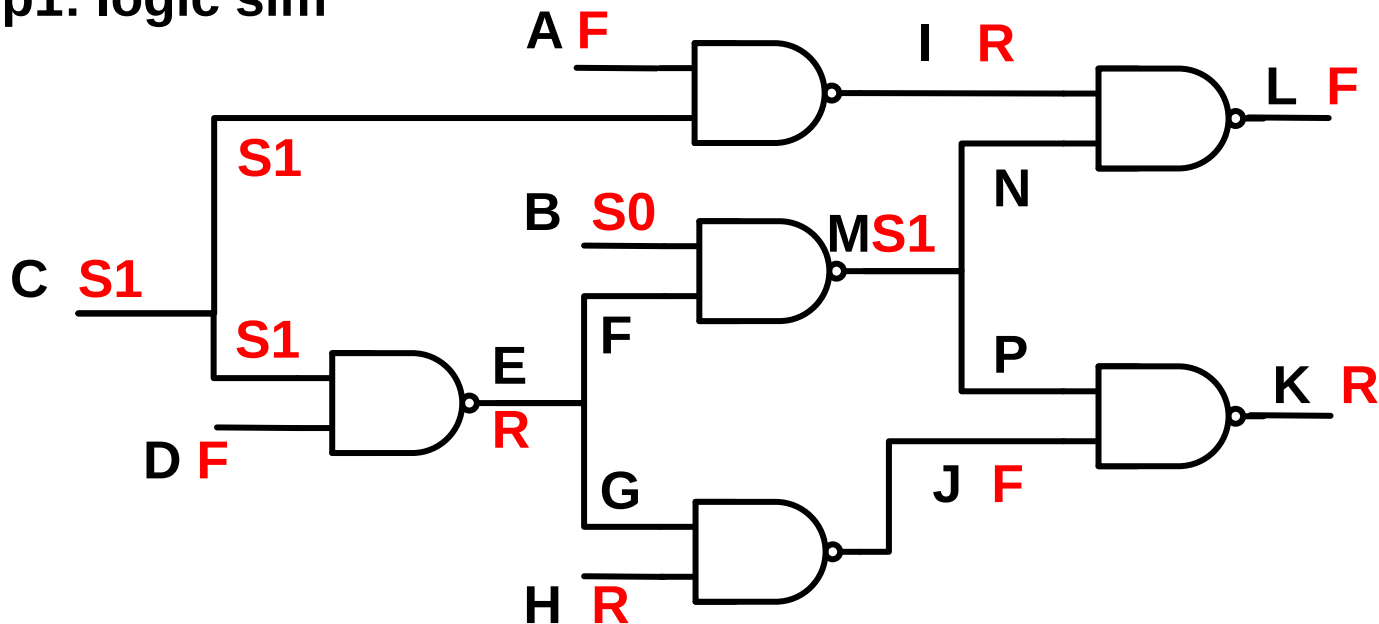
😞 More complicated algorithm

😞 Simulation fault coverage can be pessimistic (see FFT)

Number of PDF Can be WC Exponential

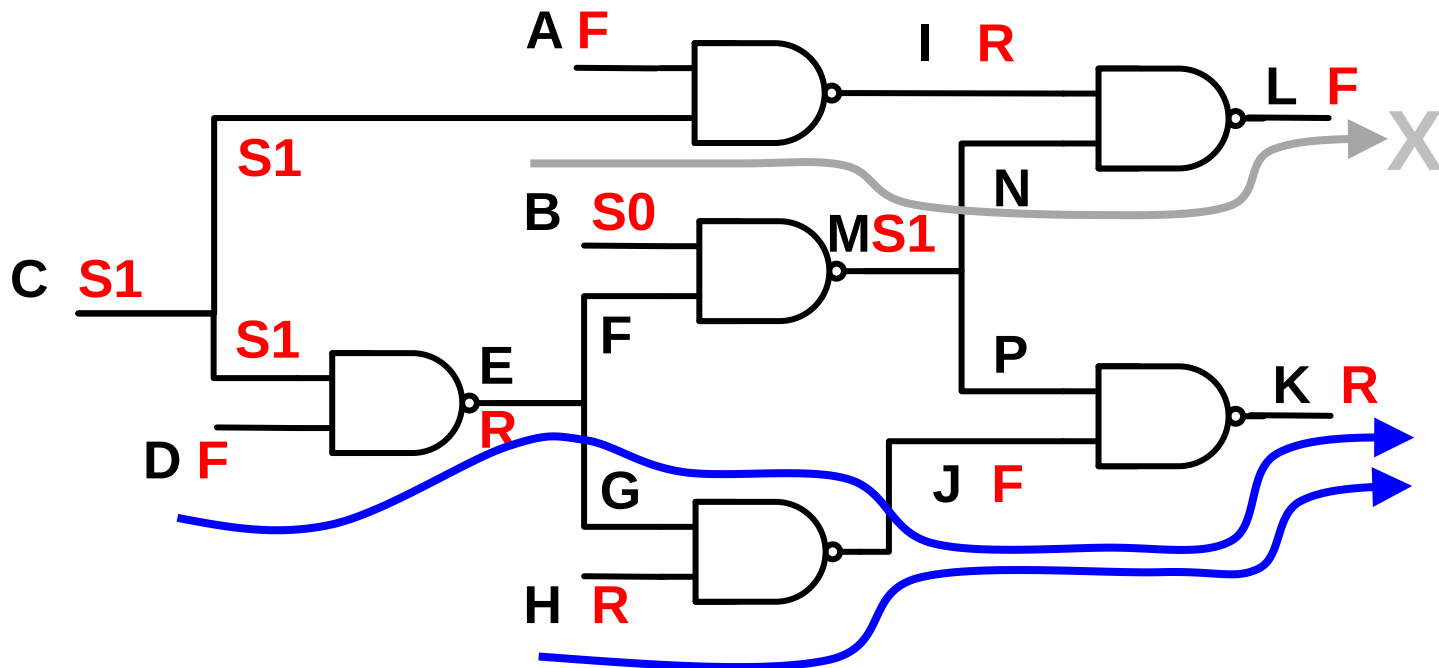
Enumerative Method [Smith 85]

- Given test set, and path delay fault list.
 1. Perform **six-valued** logic simulation
 2. Count number of sensitized paths in fault list
 3. Calculate fault coverage
- Example: ISCAS'85 C17 benchmark circuit.
 - Given test pattern and $\{\downarrow\text{DEGJK}, \downarrow\text{HJK}, \uparrow\text{BMNL}\}$. Robust FC=?
 - Step1: logic sim



Enumerative Method (2)

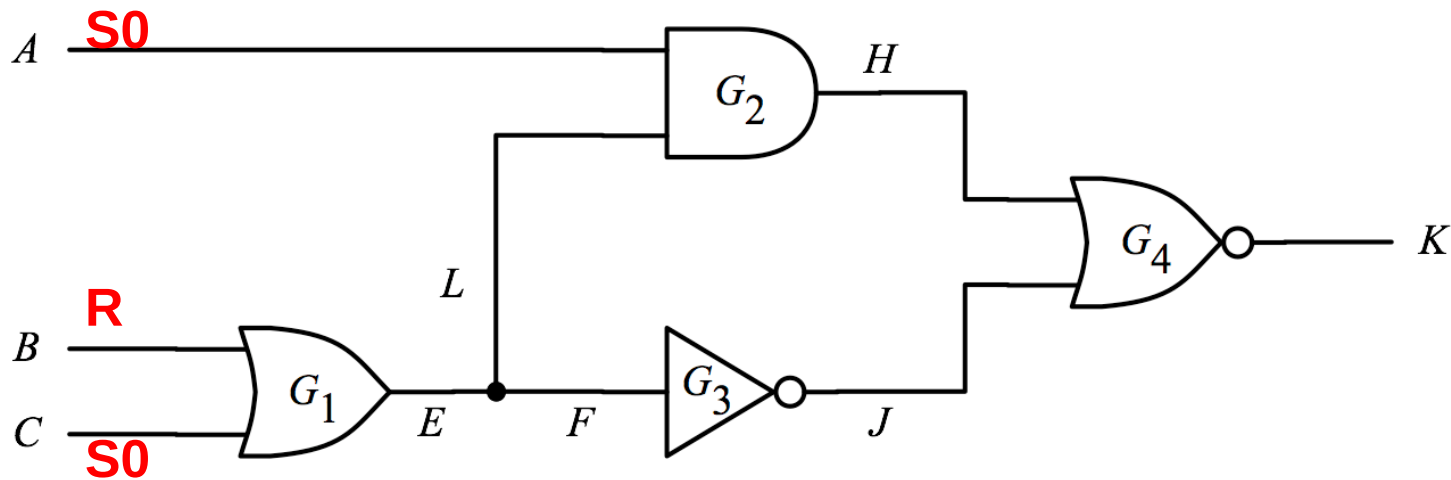
- Example: (cont'd)
 - ◆ Step2. Trace every path in fault list
 - ◆ Step3. If **every gate** on path **robustly sensitized**, PDF is detected
 - * **Detected:** $\downarrow$ DEGJK, $\downarrow$ HJK. **Not detected:** $\uparrow$ BMNL
 - ◆ Robust fault coverage = $2/3 = 66.7\%$



Quiz

Q: Apply test pattern $ABC=S0,R,S0$.
Given fault list $\{\uparrow BEFJK, \uparrow BELHK\}$
What is robust fault coverage?

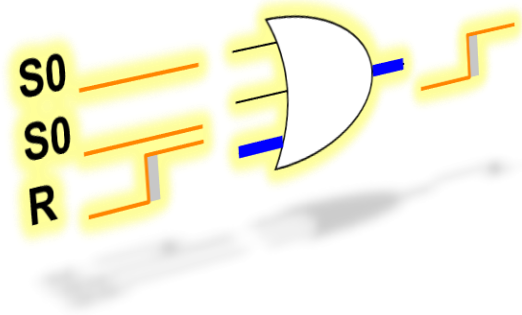
A:



But...this fault coverage is NOT representative,
since there are more than 2 PDF in this circuit.

Delay Test

- Introduction and delay fault models
- **Path Delay Fault**
 - ◆ Path Sensitization
 - ◆ Test Generation [Lin 1987]
 - ◆ Fault Simulation
 - * Enumerative method [Smith 1985]
 - * Non-enumerative method [Pomeranz 1984]
- Transition Delay Fault
- Delay Test Application
- Circuit Model for Delay Test ATPG
- Experimental Results* (not in exam)
- Issues of Delay Tests* (not in exam)
- Conclusions



Non-enumerative Method [Pomeranz 94]

Step 2,3

$$\text{Fault Coverage} = \frac{\text{number of detected faults}}{\text{number of total faults}} \times 100\%$$

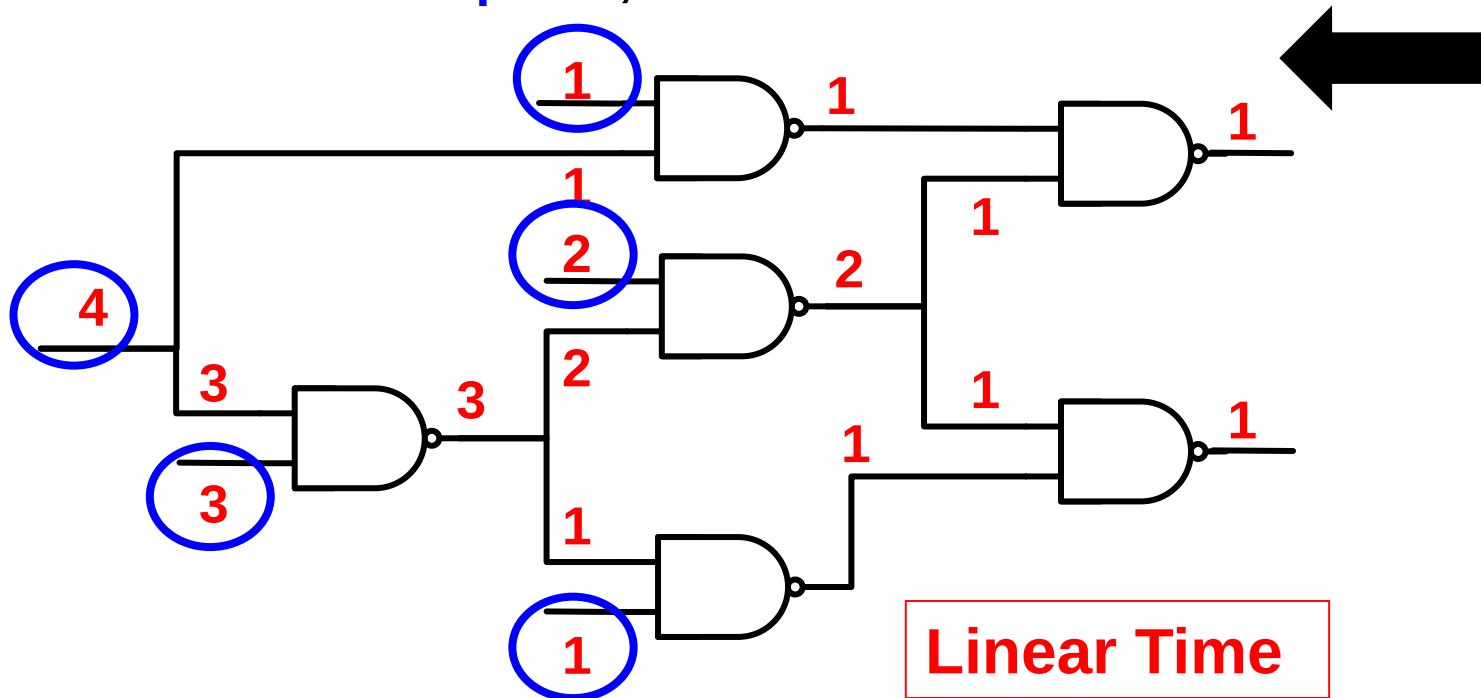
Step 1

- **Three steps**

1. Count number of total PDF
2. Count detected PDF for a test pattern
3. Count detected PDF for a test set

1. Count Number of Paths

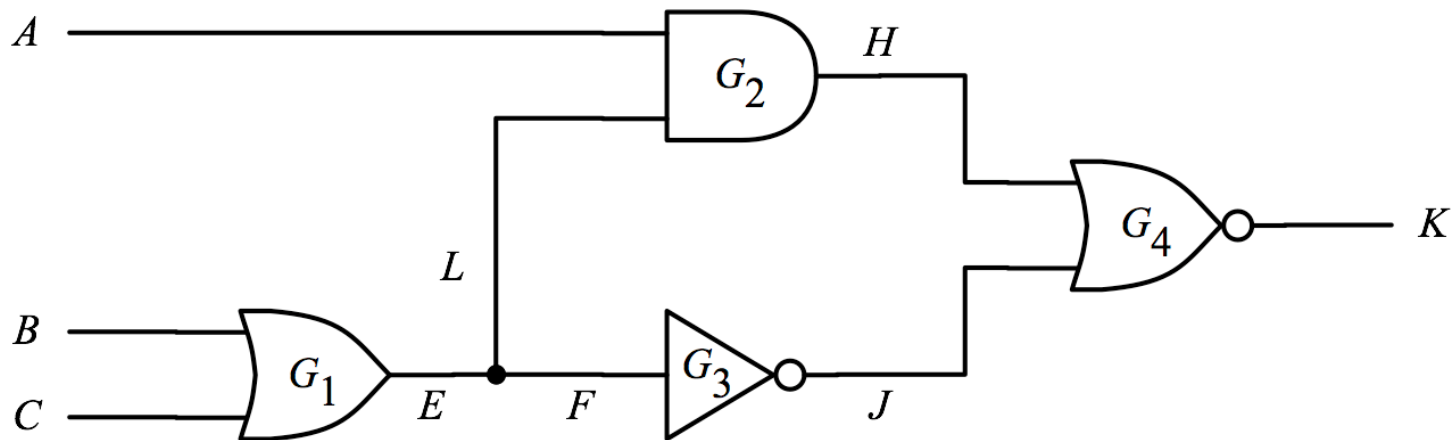
- Algorithm: Count number of paths backward, from PO to PI
 - ♦ Each PO = 1. Propagate numbers from gate output to gate inputs
 - ♦ Fanout stem = sum of branches
 - ♦ Number at PI = number of paths starting from this PI
- Example: ISCAS C17 circuit
 - ♦ Total $1+4+2+3+1=11$ paths, 22 PDF



Quiz

Q: Please count number of paths. How many path delay faults?

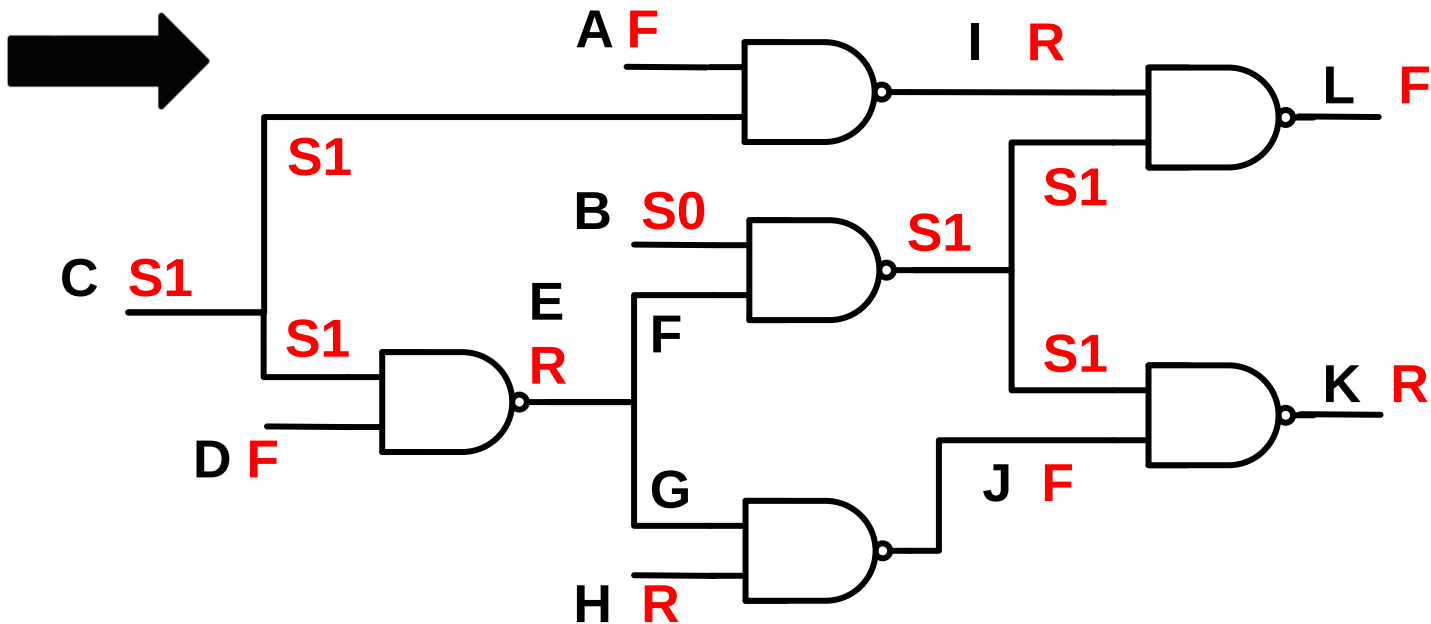
A:



2. Count Detected PDF for a Test (1)

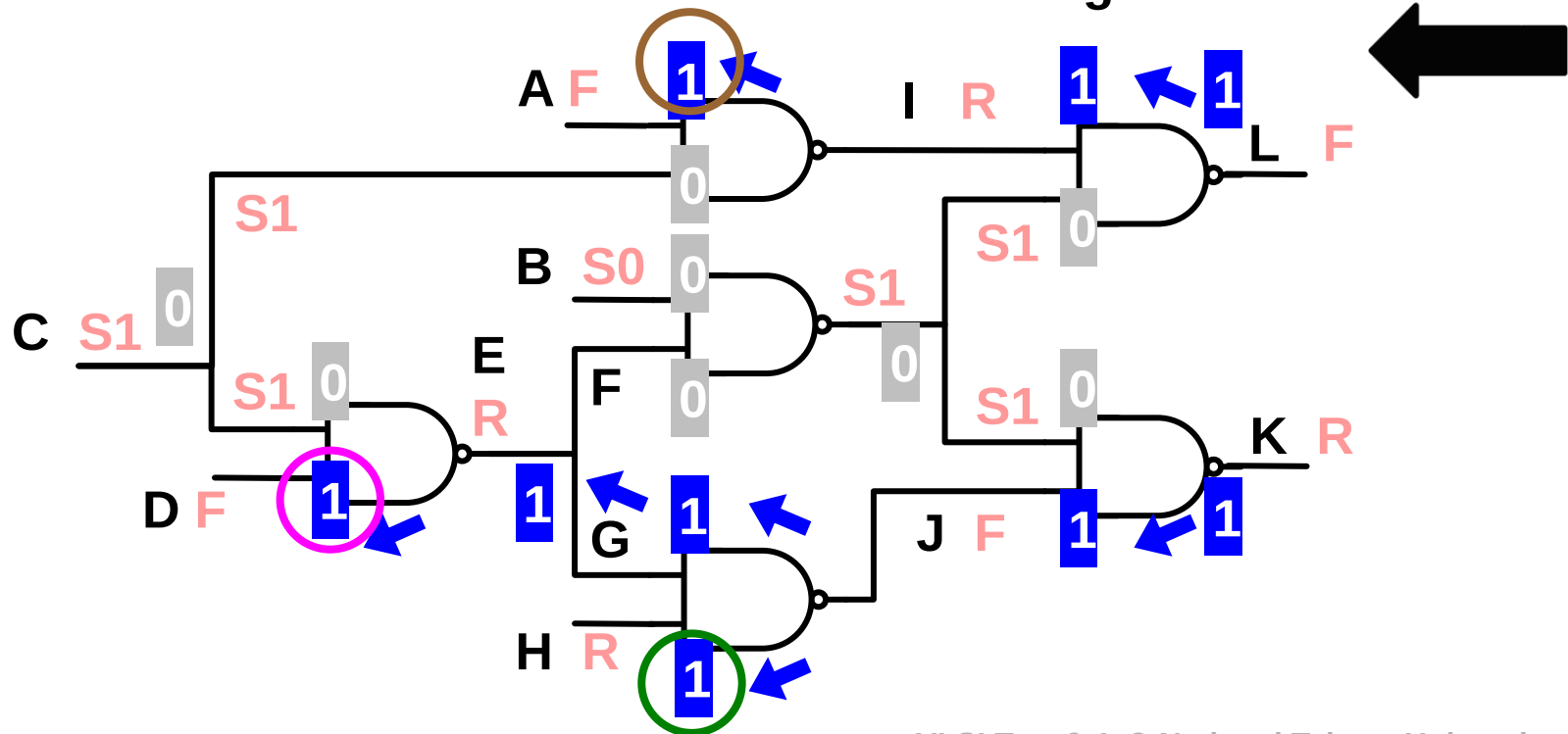
- Six-valued logic simulation, from PI to PO
- Example:
 - ♦ Apply test pattern T_1 . $A=F$, $B=S0$, $C=S1$, $D=F$, $H=R$
 - ♦ How many PDF robustly detected?

* we demo robust example here.
this technique also applicable to NR.



2. Count Detected PDF for a Test (2)

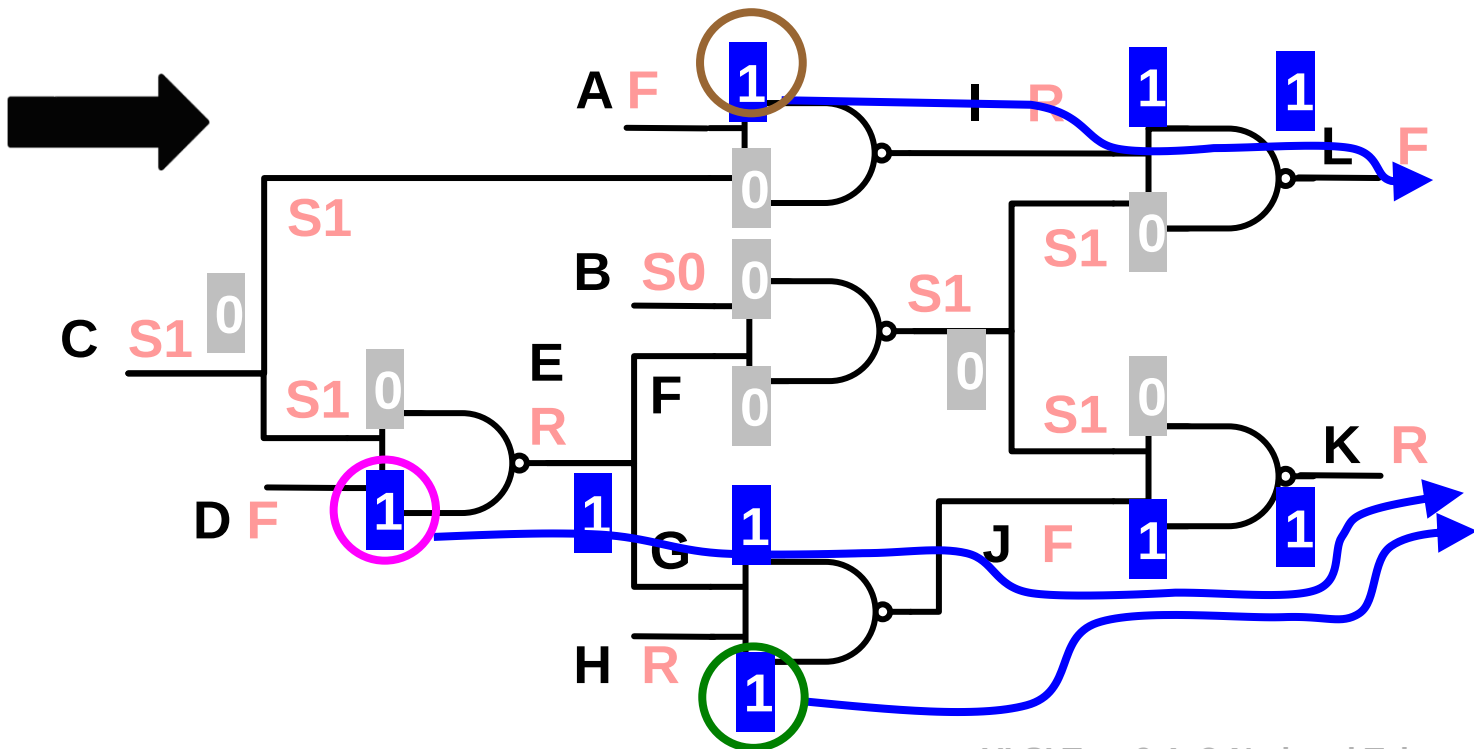
- Algorithm: Count robustly sensitized paths backward
 - ◆ If output gate is robustly sensitized, $PO=1$
 - ◆ Backward propagate numbers on robustly sensitized gate input
 - * Otherwise, number is zero
 - ◆ Fanout stem = sum of branches
 - ◆ Number at PI = number of detected PDF starting from this PI



2. Count Detected PDF for a Test (3)

- Trace non-zero numbers forward
- Test pattern T_1 robustly detects **three** PDF
 - ♦ $\downarrow$ DEGJK, $\uparrow$ HJK, $\downarrow$ AIL

* we demo robust example here.
this technique also applicable to NR.



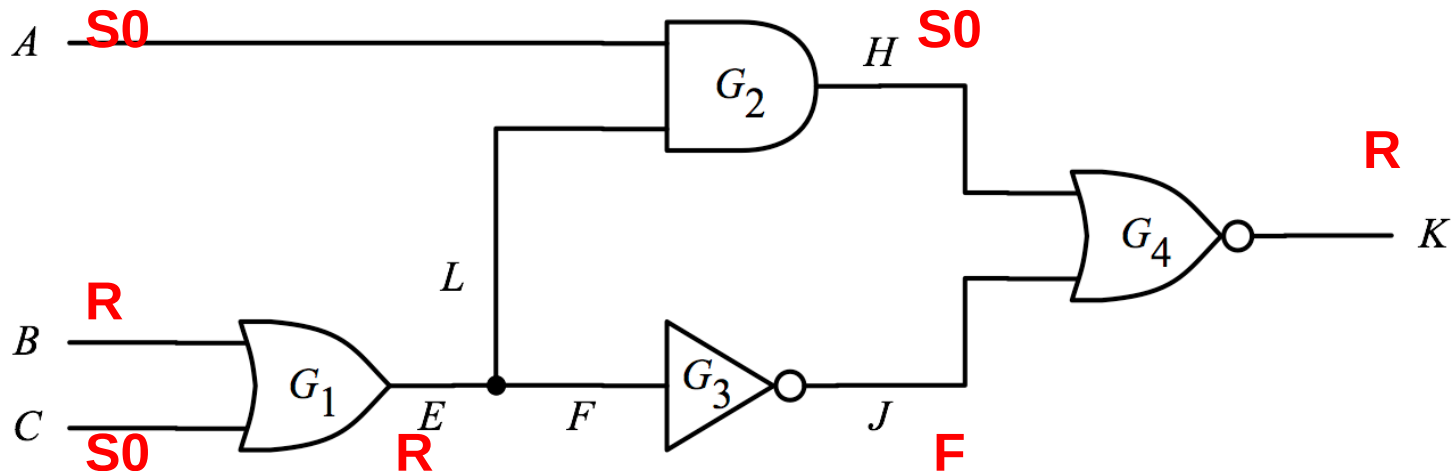
Quiz

Q: Counting backward from PO to PI. How many PDF robustly detected by this test pattern?

A:

Q: What is fault coverage of this test pattern? (Totally 10 PDF.)

A:

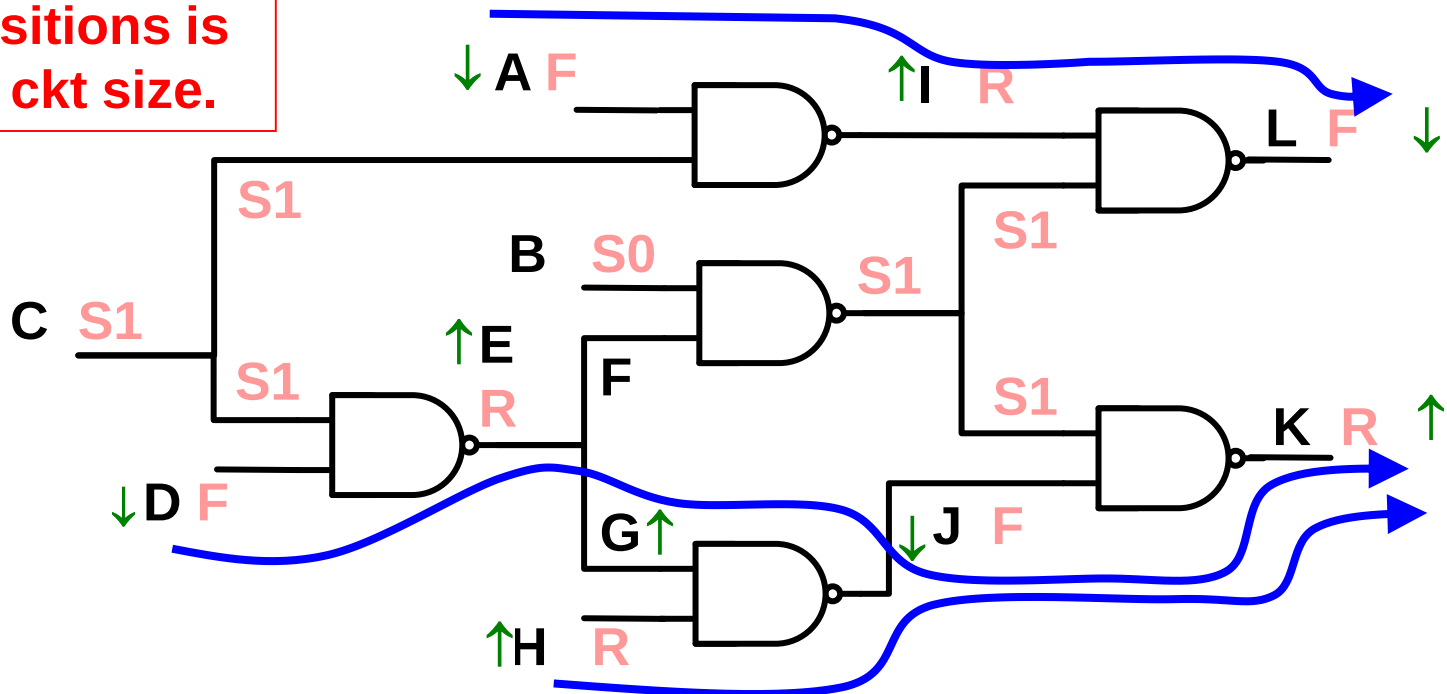


This FC is More Representative than Before.
But What If More than One Test Pattern?

What Information to Store?

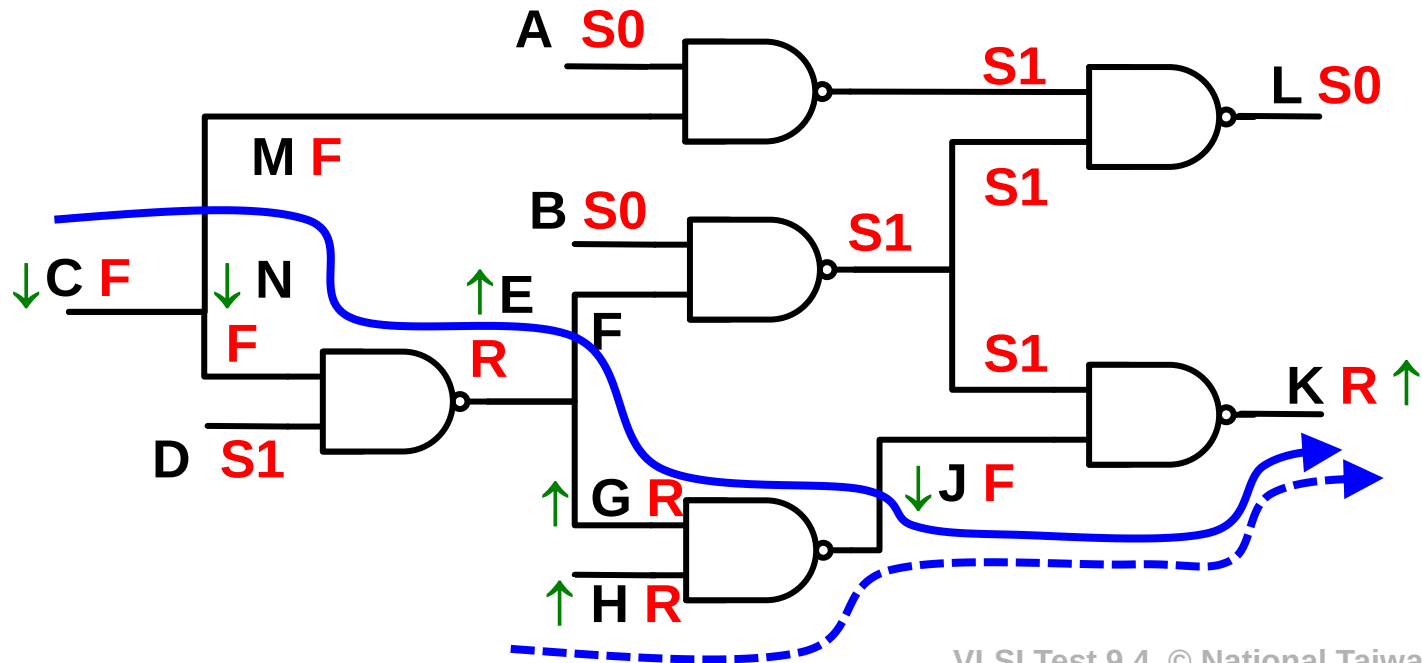
- Q: Can we store detected PDF? $\{\downarrow AIL, \downarrow DEGJK, \uparrow HJK\}$
 - ♦ A: No, because too many paths. Memory can explode!
- **Alternative: store transitions of signals on detected paths**
- Example: Store 9 transitions for T_1
 - ♦ $(A, \downarrow) (D, \downarrow) (E, \uparrow) (G, \uparrow) (H, \uparrow) (I, \uparrow) (J, \downarrow) (K, \uparrow) (L, \downarrow)$

of transitions is
linear to ckt size.



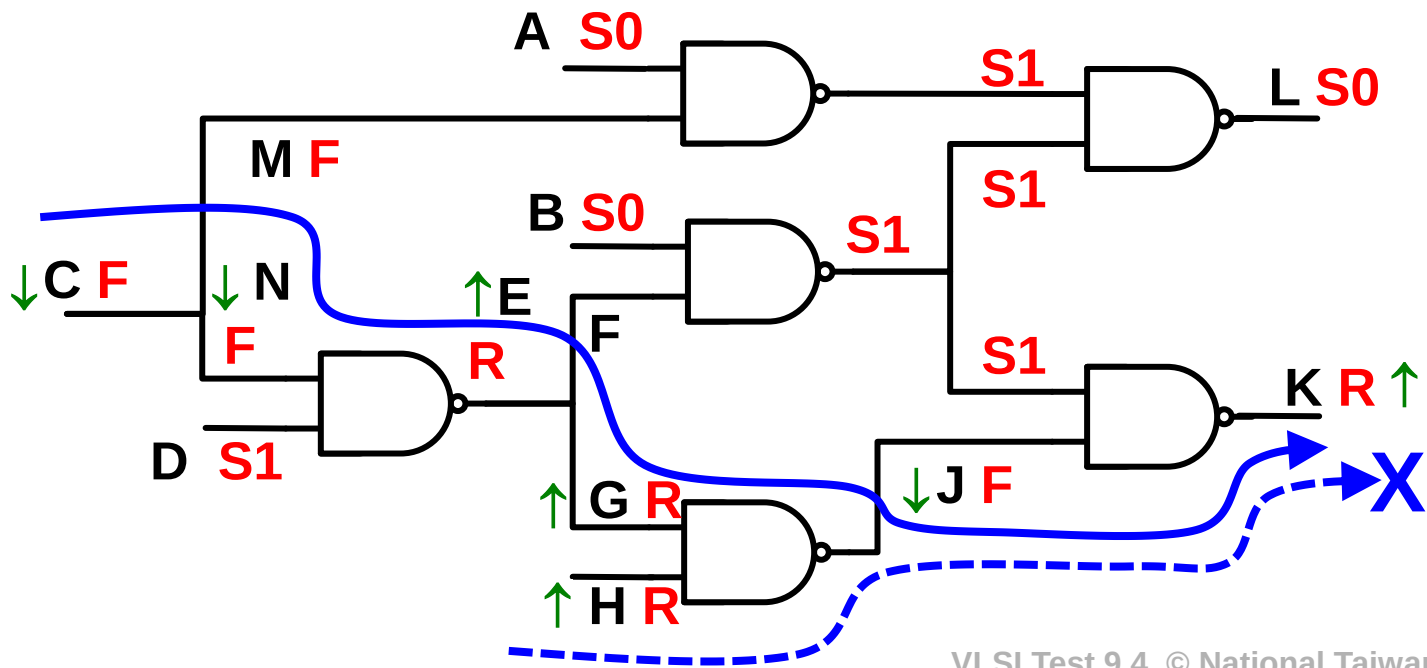
3. Count Detected PDF for a Test Set (1)

- Example: After T_1 , apply another test pattern T_2
 - ♦ Repeat step 2, T_2 detects PDF: $\{\downarrow CNEGJK, \uparrow HJK\}$
 - ♦ Transitions of T_2 : $(C, \downarrow) (N, \downarrow) (G, \uparrow) (H, \uparrow) (J, \downarrow) (K, \uparrow)$
- For test set $\{T_1, T_2\}$, fault coverage = $(3+2)/22 = 22.7\%$?
 - ♦ No! $\uparrow HJK$ was detected by T_1 . Cannot count twice
 - ♦ But we didn't store PDF. **How do we know $\uparrow HJK$ was detected?**



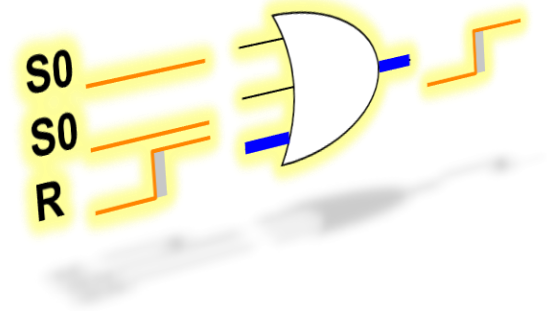
3. Count Detected PDF for a Test Set (2)

- Criterion: To count PDF as **newly detected**,
 - ♦ Require **at least one transition did NOT appear previously**
- $\downarrow\text{CNEGJK}$ is newly detected by T_2
- $\uparrow\text{HJK}$ is **NOT newly detected** because all transition appeared in T_1
 - ♦ Transitions of T_1 : $(A,\downarrow) (D,\downarrow) (E,\uparrow)(G,\uparrow)(H,\uparrow)(I,\uparrow)(J,\downarrow)(K,\uparrow)(L,\downarrow)$
 - ♦ Transitions of T_2 : $(C,\downarrow) (N,\downarrow) (G,\uparrow)(H,\uparrow)(J,\downarrow)(K,\uparrow)$
- For test set $\{T_1 T_2\}$, robust fault coverage = $(3+1) / 22 = 18.1\%$



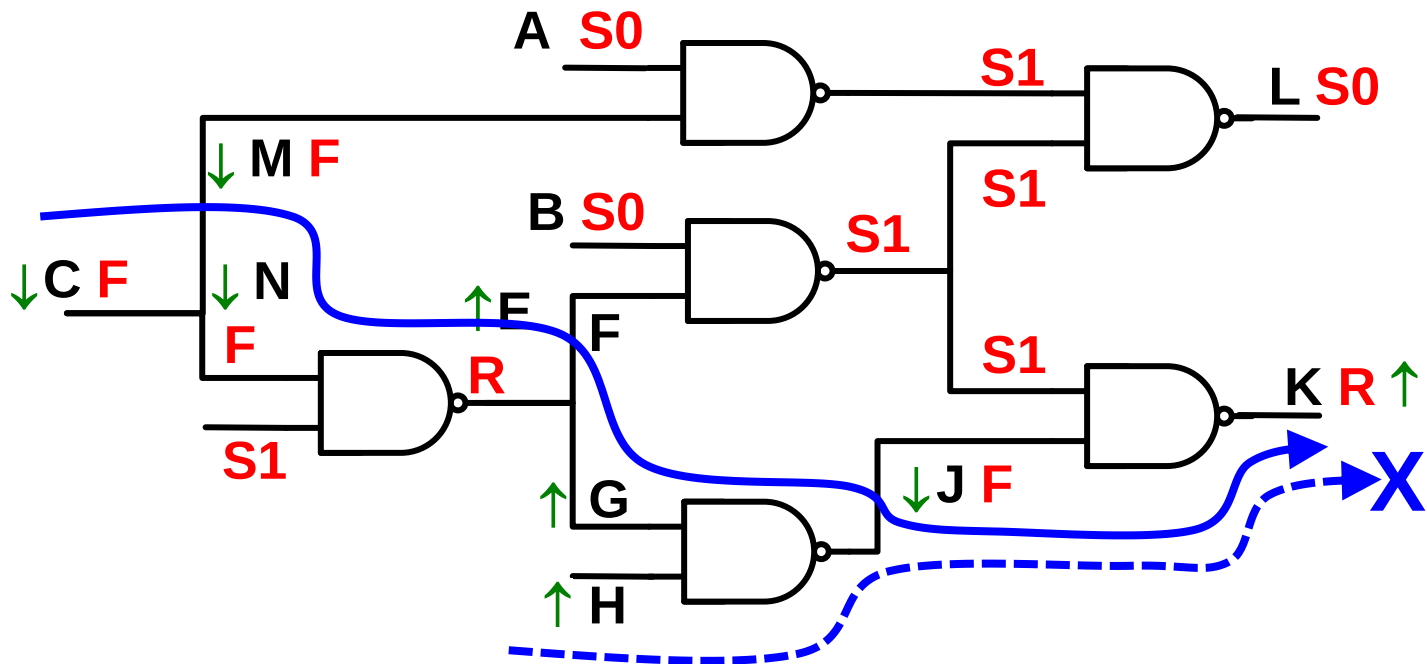
Summary

- Logic system depends on requirement
 - ♦ **Five-valued** logic for robust test pattern generation
 - ♦ **Six-valued** logic for fault simulation (without unknowns)
- **Enumerative** method
 - ♦ Users need to provide fault list
- **Non-enumerative** method
 - ♦ Users don't need to provide fault list
 1. Counting number of total PDF
 2. Counting number of detected PDF for a test pattern
 3. Counting number of detected PDF for a test set
 - ♦ Store **transitions**, instead of PDF
 - * Memory usage **won't go exponential**



FFT

- Q: Storing transitions, instead of PDF, can result in **pessimistic FC**
 - ♦ Simulation FC can be **lower than** actual FC
 - ♦ Why? please show example



FFT2

- Q: In this video, we use robust test as examples.
 - ♦ Can we apply 6-valued logic to simulate **non-robust** path delay fault coverage? (given fully specified test patterns)

